

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain Microcontroller design and specification of medical field.
- Q.37 Write a note on various types of interrupts used in Microcontroller?
- Q.38 Write a note on all Arithmetic and logical operations used in 8051?

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4th Semester/ Med. Eltx.
Subject : Microcontrollers Applications in Medical Technology

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 LCD stands for :
- a) Liquid Cristal Display b) Liquid code display
c) Both d) None of these
- Q.2 Nibble consists of:
- a) 4 bit b) 1 bit
c) 0 bit d) None of these
- Q.3 Two reverse terms are:
- a) Bit to bit converter b) A/D & D/A converter
c) Both d) None of these
- Q.4 Operation for 5v power supply is:
- a) Least 1 b) Lost 1
c) Logic 1 d) None of these
- Q.5 Main operations are :
- a) Reset b) Hold
c) Both d) None of these

- Q.6 Interrupts in microcontroller 8051 can be :
 a) 10 b) 5
 c) 1000 d) 60
- Q.7 LCD stands for
 a) Light Code Do b) Least Code Do
 c) Light Crystal Display d) None of these
- Q.8 Microcontroller 8051 consists of
 a) Mnemonics b) Hexadecimal code
 c) Combination of both d) None of these
- Q.9 Microcontroller 8051 also known as :
 a) MCS-51 b) MS-51
 c) SM-51 d) None of these
- Q.10 Applications of Microcontroller are:
 a) Robotics b) Embedded System
 c) Medical application d) All of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Write one application of 8031 Microcontroller.
- Q.12 Write pin number of Input and Output port in 8051 Microcontroller?
- Q.13 How many Addressing modes are used in 8051 Microcontroller?
- Q.14 For disabling the interrupts _____ register is set to zero.
- Q.15 Define 7-Sement interface.
- Q.16 How many instructions used in 8051?

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- Q.17 RTC stands for _____.
- Q.18 What is embedded system?
- Q.19 Define Compiler.
- Q.20 What do you mean by bug?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define role of pin 1 to 8 of Microcontroller 8051.
- Q.22 Define Accumulator and its role for 8051.
- Q.23 Define PSW in detail inside 8051.
- Q.24 How PIC is different from another microcontroller?
- Q.25 What is DMA controller?
- Q.26 Write 5 similarities and differences between 8051 and 8031?
- Q.27 Define Immediate Addressing mode.
- Q.28 Define Direct Addressing mode.
- Q.29 Write a note on Timer flag interrupts?
- Q.30 Write a note on modes of timer in microcontroller.
- Q.31 Write a note on internal and external interrupts?
- Q.32 Write 5 features of 8051?
- Q.33 Define role of pin 21-28 of Microcontroller 8051?
- Q.34 Why PIC microcontroller used in medical field?
- Q.35 Write a note on pin 32-39 of Microcontroller 8051?

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